

E-Commerce Sales Analysis Dashboard

Project Report

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Tool: Power BI, Power Query, DAX, Excel, SQL, Python

1. Abstract

This project analyzes e-commerce sales data and presents key business insights through an interactive Power BI dashboard. The dashboard helps decision-makers monitor sales performance, profitability, orders, product performance, and category-wise trends using interactive filters.

2. Objectives

- Analyze overall sales and profit performance.
- Track monthly sales and profit trends.
- Identify top-selling products.
- Compare sales across product categories.
- Build an interactive dashboard using Power BI.

3. Dataset

Dataset: Sample Superstore (E-Commerce Sales Dataset)

Main Columns:

- Order Date
- Product Name
- Category
- Region
- Sales
- Profit
- Quantity
- Discount
- Customer Name

4. Tools & Technologies

- Microsoft Power BI
- Power Query
- DAX
- Microsoft Excel

5. Data Preparation

- Imported dataset into Power BI.
- Removed duplicate records.
- Checked missing values.
- Corrected data types.
- Created date hierarchy.
- Built calculated measures using DAX.

6. Dashboard Components

KPI Cards:

- Total Sales
- Total Profit
- Total Orders
- Profit Margin %

Visuals:

- Sales by Month
- Profit by Month
- Top 5 Selling Products
- Sales by Category

Filters:

- Year
- Month
- Category
- Region

7. DAX Measures

Total Sales = SUM(Sales)

Total Profit = SUM(Profit)

Total Orders = DISTINCTCOUNT(Order ID)

Profit Margin % = DIVIDE([Total Profit],[Total Sales],0)

8. Key Insights

- Total Sales reached approximately 2M.
- Total Profit reached approximately 286K.
- Profit Margin is around 12.47%.
- November recorded the highest monthly sales.
- Technology generated the highest category sales.
- Canon products contributed the highest product sales.

9. Business Recommendations

- Increase inventory before peak sales months.
- Promote top-selling products.
- Focus on high-performing categories.
- Review low-profit periods for cost optimization.
- Improve pricing and discount strategies.

10. Conclusion

The dashboard provides a clear view of business performance and supports data-driven decision-making through interactive reports and KPI monitoring. It can be extended with forecasting, customer segmentation, and regional performance analysis.